

A From-Scratch CUDA Forward-Pass Inference Engine for a LeNet-Style MNIST CNN

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Abstract—I build a forward-pass inference engine for a LeNet-style MNIST convolutional network in C++ and CUDA, with no deep-learning framework at runtime. Five kernels — 2D convolution, max pooling, ReLU, linear, and log-softmax — are written first as naive reference implementations, then convolution and the linear layer are re-written as tiled shared-memory variants. A three-layer correctness harness validates each kernel against a CPU reference and the composed pipeline against PyTorch on five MNIST fixtures. On an NVIDIA RTX 4090, the full GPU forward pass agrees with PyTorch to a maximum absolute error of 1.91×10^{-5} over the entire 10 000-image MNIST test set, with 97.93% accuracy and 100% argmax agreement. Per-kernel benchmarks show tiled GEMM reaching $6.43\times$ over naive at $1024 \times 1024 \times 1024$; tiled convolution slightly regresses at LeNet scale because the small working set fits in L1, making tiling’s synchronization cost dominant. I also implemented a cuDNN baseline to see how the direct kernels compare to the vendor library, which turns out to be slower at LeNet shapes due to descriptor and dispatch overhead, and only pulls ahead on a larger $32 \rightarrow 64$, 64×64 synthetic workload.

Index Terms—CUDA, GPU computing, convolutional neural networks, MNIST, inference, tiled matrix multiplication, shared memory.

I. INTRODUCTION

Deep-learning frameworks hide the GPU primitives behind autograd, graph capture, and vendor libraries. This project builds the inference path of a small convolutional network from the CUDA kernels up, with no framework code and no vendor-provided neural-network primitives at runtime. The target network is a LeNet-style MNIST classifier from my prior project work (two conv layers, two max pool layers, two fully connected layers, ReLU activations, and a log-softmax output). Convolution is the right primitive here because images have spatial structure: nearby pixels are correlated, useful features like edges and textures appear at multiple locations, and a convolutional filter can detect them anywhere in the image without learning a separate weight for each position. The correctness target is matching a PyTorch forward pass on the MNIST test inputs to within 1×10^{-4} absolute error per log-softmax element.

All model weights are exported once from the trained PyTorch checkpoint as header-less raw float32 binaries and copied to the GPU via `cudaMemcpy`; PyTorch is never linked at runtime. The work covers naive CUDA kernels for all five operators, tiled shared-memory variants for convolution

and the linear layer, and a three-layer correctness harness that validates each kernel against a CPU reference before it enters the full pipeline. Beyond the core scope, I also implemented a fused conv+bias+ReLU+pool kernel that eliminates two intermediate DRAM round-trips per conv stage, a cuDNN baseline that required explicit algorithm and precision-mode configuration to produce numerically comparable results (Section IV-E), and a compiler-level occupancy analysis characterizing register, shared-memory, and warp utilization for all eight kernels (Section IV-D). Each produced a finding that was not obvious before the work was done.

II. RELATED WORK

The target network follows the LeNet-5 design of LeCun *et al.* [2], which introduced the convolution, subsampling, and fully connected layer structure used throughout this project and established MNIST as the canonical digit-recognition benchmark. While LeNet predates the GPU-accelerated training boom, its small parameter count (approximately 21 800 weights in the variant used here) makes it a practical fit for a from-scratch forward-pass implementation: the full weight set fits in a few tens of kilobytes and each kernel shape is small enough that hand-computed references can be built for unit testing.

Krizhevsky, Sutskever, and Hinton [1] made the practical case for GPU-accelerated CNN inference: their AlexNet implementation showed that hand-written CUDA convolution and fully-connected kernels on a pair of GTX 580s could achieve top-5 ImageNet accuracy far beyond the then-state-of-the-art, directly motivating the per-kernel CUDA approach taken here.

Two papers on the GPU-kernel side inform the tiled kernels. Volkov and Demmel [4] characterized GPU memory bandwidth and arithmetic throughput and demonstrated how shared-memory tiling combined with register blocking approaches peak machine performance for dense linear algebra; this motivates the tiled GEMM used for the fully connected layers. Lavin and Gray [3] reduced convolution arithmetic with Winograd minimal-filter algorithms, showing the gap that remains between direct-convolution implementations (the approach taken in this project) and algorithm-level reformulations; this is context for interpreting the gap between the direct kernels and a cuDNN baseline, which selects Winograd

or implicit-GEMM algorithms per shape. Chetlur *et al.* [5] describe cuDNN itself, used here as the throughput baseline.

III. METHODS

A. Target Network and Weight Export

The target network is the LeNet-style model from my CS5330 Project 5, summarized in Fig. 1: conv1 (1→10, 5×5, stride 1), max-pool 2×2, ReLU; conv2 (10→20, 5×5, stride 1), Dropout2d (skipped at inference), max-pool 2×2, ReLU; flatten to 320, fc1 (320→50), ReLU, fc2 (50→10), log-softmax. The two convolutional stages learn a hierarchy of spatial features — conv1 learning low-level detectors such as edges and stroke orientations, conv2 combining those into higher-level digit-part representations — before the fully connected layers map that representation to class scores. The eight learnable tensors (conv1/conv2/fc1/fc2, each with weight and bias) are exported once from the trained .pth checkpoint by a short Python tool (tools/export_weights.py) as headerless raw float32 binaries in PyTorch’s native memory layout (NCHW for conv filters, (out, in) for linear weights). At inference, the host loader reads these files with a plain fread and cudaMemcpy the contents to the device, so Python and PyTorch are not runtime dependencies of the engine.

B. Correctness Harness

Correctness is built up in three layers, each designed to catch a different class of bug before it can reach the next. The first layer is a CPU reference implementation (host/reference.cpp), written in plain C++17 with no tuning and no parallelism, covering every operator in the network. This reference is validated against five PyTorch log-softmax fixtures captured from the trained model on MNIST test indices 0–4; the maximum absolute error is on the order of 4×10^{-6} , which is the noise floor from different float32 summation orders between PyTorch and my reference loop. Every CUDA kernel is then diffed against this reference.

The second layer is a per-kernel CUDA diff: each naive kernel (conv2d, max_pool2d, ReLU, linear, log_softmax) ships with a unit test that generates random mixed-sign inputs at several shapes — LeNet’s own shapes plus deliberately off-model shapes with odd dimensions — and diffs the kernel’s output element-wise against the CPU reference run on the same inputs. The tolerance is the project-wide 1×10^{-4} . The third layer is the end-to-end GPU forward pass: all five naive kernels are composed through device-pointer launchers with activations kept on-device between layers, and the full pipeline is run on the same five PyTorch fixtures used to validate the CPU reference. This exercises not just each kernel but also the integration glue — tensor layouts, layer ordering, and the handling of the flatten between conv2/pool/relu and fc1.

C. Toolchain and Hardware

All CUDA development and benchmarks run on a single RunPod instance: NVIDIA GeForce RTX 4090 (24 GB

GDDR6X, compute capability 8.9), Ubuntu 22.04, NVIDIA driver 550.127.05, CUDA Toolkit 12.4 (nvcc V12.4.131), g++ 11.4.0, CMake 3.22.1. Builds are -O3 (Release). The CMakeLists.txt targets compute capabilities 75/80/86/89, so kernels remain portable to T4, A100, and 30-/40-series GPUs.

D. Naive Kernels

Each operator is implemented first as a direct translation of the mathematical definition, with one thread per output element and no shared-memory reuse. The priority at this stage is readability and correctness, not performance; every naive kernel passes its unit test at the float32 noise floor (see Table I).

Convolution uses one thread per output pixel. The grid is launched as $(\lceil W_{\text{out}}/16 \rceil, \lceil H_{\text{out}}/16 \rceil, N \cdot C_{\text{out}})$ with 16×16 thread blocks. Each thread loops over $C_{\text{in}} \cdot K_h \cdot K_w$ weight/input pairs and accumulates the dot product in a register; the bias is added at the end. Tensors use PyTorch-native NCHW layout throughout. Stride is 1 and padding is 0, matching the Project 5 network.

Max pooling launches one thread per output pixel over a 2×2 non-overlapping window with stride 2. Each thread loads four input values, takes the max, and writes one output; no reduction primitive is required at this window size.

ReLU is a trivial elementwise kernel with a 1D grid: each thread reads one element, clamps it at zero, and writes it back. It exists as a separate kernel rather than being fused because the naive-first design discipline treats each operator as an independent testable unit.

Linear maps to a GEMM of shape $(N, \text{in}) \times (\text{in}, \text{out})$; the naive kernel launches one thread per output element of the (N, out) matrix, with each thread computing a dot product over in_features weight/input pairs plus a bias.

Log-softmax is the only kernel with a non-trivial numerical trap. A straightforward implementation of $\log(e^{x_i} / \sum_j e^{x_j})$ overflows for logits with magnitude larger than ~ 88 because e^x exceeds the float32 range. I therefore subtract the per-row maximum before exponentiation: $\text{logsoftmax}(x)_i = (x_i - m) - \log \sum_j e^{x_j - m}$ where $m = \max_j x_j$. Each thread block handles one row. `blockDim.x` is set to the next power of two greater than or equal to the class count, which lets both the max-reduction and the sum-of-exp reduction use a clean shared-memory tree-reduction with sentinel padding ($-\infty$ for max, 0 for sum) in the out-of-range lanes. The in-device intrinsics `__expf` and `__logf` are used for the elementwise exponentials and final log.

E. Tiled Kernels

Two operators are re-implemented as shared-memory tiled variants: convolution and the linear layer. Both are in separate files from the naive kernels and go through the same CPU-reference validation at the same tolerance before being wired into the end-to-end pipeline.

The tiled convolution block produces one `TILE_H` $\times$ `TILE_W` output patch for one (n, oc) pair, with `TILE_H` =

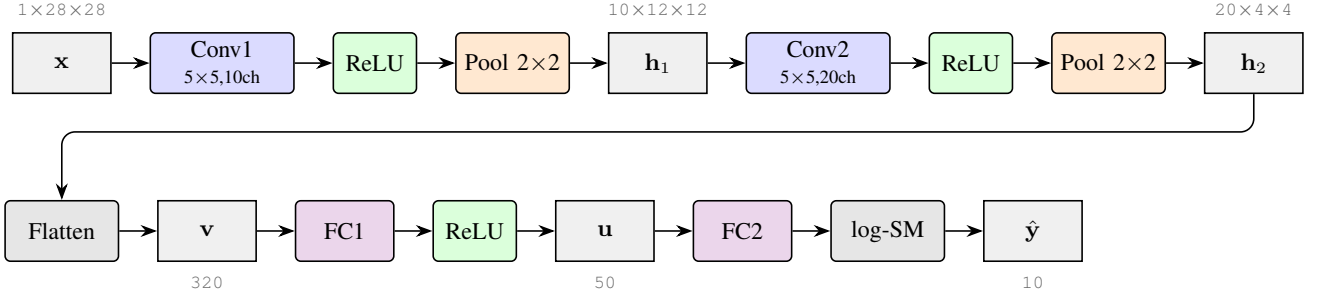


Fig. 1. LeNet-MNIST forward-pass pipeline as implemented by this engine. Six named tensor stages — input $\mathbf{x}$, two convolutional activation maps $\mathbf{h}_{1,2}$, flattened vector $\mathbf{v}$, hidden vector $\mathbf{u}$, output log-probabilities $\hat{\mathbf{y}}$ — are connected by ten separate CUDA kernel launches. Transitions in the figure aggregate the kernels that operate on the same feature-map shape; the underlying launch sequence is `conv1`, `max_pool2d`, `ReLU`, `conv2`, `max_pool2d`, `ReLU`, `flatten` (an implicit reshape, not a launched kernel), `linear (fc1)`, `ReLU`, `linear (fc2)`, `log_softmax`. Activations remain on-device between kernels: only $\mathbf{x}$ is copied host-to-device and only $\hat{\mathbf{y}}$ is copied back.

$\text{TILE_W} = 16$. Dynamic shared memory holds a $(\text{TILE_H} + K_h - 1) \times (\text{TILE_W} + K_w - 1)$ input patch for one input channel at a time. The threads in the block cooperatively load that patch (with zero-padding for the halo that falls outside the input), then each in-bounds thread accumulates its output pixel’s contribution over all $K_h \cdot K_w$ weight elements. A `__syncthreads` sits between the load and the accumulation, and a second `__syncthreads` at the end of the input-channel iteration guards shared memory before the next C_{in} step overwrites it. The weight tensor is read directly from global memory; at LeNet scale the full weight tensor is small enough (5000 floats for `conv2`) to reside in L1 after a few iterations, so explicit caching of weights was not pursued.

The tiled GEMM follows the standard Volkov-style block-tile structure [4]. Each block computes a $\text{BM} \times \text{BN}$ output tile; input tiles A_s (a $\text{BM} \times \text{BK}$ slab of the input batch) and B_s (a $\text{BK} \times \text{BN}$ slab of the weight matrix) are loaded into shared memory, a per-thread register accumulator runs the inner BK -long dot product with `#pragma unroll`, and then the next K -panel is loaded. Boundary handling uses zero padding on out-of-range loads so the inner loop has no branches. The block dimensions are $\text{BM} = \text{BN} = \text{BK} = 16$; the weight matrix is loaded in its natural (out, in) layout and transposed implicitly at load time into B_s .

IV. EXPERIMENTS AND RESULTS

A. Numerical Parity

Table I summarizes the correctness results collected as each kernel landed. All per-kernel tests pass the project-wide $1e-4$ absolute-error tolerance by several orders of magnitude; observed errors are at the float32 summation-reorder noise floor. The full CPU reference forward pass matches PyTorch log-softmax on all five fixtures, establishing the oracle used for subsequent GPU kernels. The end-to-end CUDA forward pass — all five naive kernels composed through device-pointer launchers with activations held on-device between layers — matches PyTorch log-softmax on every fixture with an overall maximum absolute error of $5.72e-6$ (roughly four orders of magnitude inside the tolerance), and the predicted argmax agrees with PyTorch on every image.

To validate the engine beyond the five hand-picked fixtures, a full-test-set harness (`test_forward_mnist10k.cu`) runs the pipeline against all 10 000 MNIST test images under PyTorch log-softmax as the oracle. Across 100 000 logits, the maximum absolute error is $1.91e-5$ for both the naive and the tiled pipelines; the argmax agrees with PyTorch on every image; and the engine’s top-1 accuracy against the ground-truth labels (97.93%) is identical to PyTorch’s eval-mode accuracy on the same weights. Human error on MNIST is approximately 0.2%; 97.93% is consistent with published results for LeNet-5 variants at this parameter scale [2].

B. Throughput

All timings below are CUDA-event measurements collected by `tools/bench.cu`: each kernel is invoked with a handful of untimed warmup iterations, then repeated 200 times (or 50 times for the end-to-end forward pass at batch 64) between a start and stop `cudaEvent`, and the reported value is the mean per-call time in milliseconds. Inputs are allocated and populated with random values once per measurement. Device-pointer variants of each kernel are used so that `cudaMalloc/cudaMemcpy` cost is not charged against the per-kernel number. The end-to-end forward-pass number includes the per-call host-to-device copy of the full weight set and input batch plus the device-to-host copy of the output logits, but not the one-time load of weight files from disk.

Table II compares the naive and tiled convolution kernels at the two LeNet conv shapes, at a batch-64 variant of each, and at an off-model synthetic shape ($32 \rightarrow 64$, 5×5 filter on a 64×64 input) chosen to give the tile more work per output block. Table III does the same for the linear kernel, including a $1024 \times 1024 \times 1024$ synthetic case to exercise the tiled GEMM at a problem size where shared-memory reuse actually dominates. Table IV reports end-to-end forward-pass time for the full LeNet pipeline at batch 1 and batch 64.

The tiled convolution kernel is a slight regression at every LeNet-scale shape. The naive kernel issues $C_{\text{in}} \cdot K_h \cdot K_w = 25$ or 250 global loads per output thread depending on the layer; the weight tensor is 250 floats for `conv1` and 5000 for `conv2`, small enough that after the first few output blocks

TABLE I

PER-TEST MAXIMUM ABSOLUTE ERROR VS. THE APPLICABLE REFERENCE (PYTORCH LOG-SOFTMAX FOR THE CPU FORWARD PASS; THE CPU REFERENCE FOR EACH CUDA KERNEL). TOLERANCE: 1×10^{-4} .

Test	Shape / input	Max abs err
CPU fwd vs PyTorch	fixture 0 (MNIST idx 0)	$\sim 4e-6$
CPU fwd vs PyTorch	fixtures 1-4	$\leq 4e-6$
conv2d naive	$N=2, 1 \rightarrow 10, 5 \times 5 @ 28 \times 28$	$1.43e-6$
conv2d naive	$N=2, 10 \rightarrow 20, 5 \times 5 @ 12 \times 12$	$3.82e-6$
conv2d naive	odd-dim $3 \rightarrow 5, 3 \times 3 @ 13 \times 17$	$9.54e-7$
max_pool2d naive	$N=2, C=10, 24 \times 24$	$0.00e+0$
max_pool2d naive	$N=2, C=20, 8 \times 8$	$0.00e+0$
max_pool2d naive	odd-dim $C=3, 5 \times 7$	$0.00e+0$
relu naive	post-conv1 pool (1440)	$0.00e+0$
relu naive	post-conv2 pool (320)	$0.00e+0$
relu naive	fc1 output (50)	$0.00e+0$
relu naive	odd 1023	$0.00e+0$
linear naive	LeNet fc1 ($N=2, 320 \rightarrow 50$)	$2.86e-6$
linear naive	LeNet fc2 ($N=8, 50 \rightarrow 10$)	$9.54e-7$
linear naive	off-model ($N=3, 1024 \rightarrow 17$)	$7.63e-6$
log_softmax naive	LeNet final ($N=1, C=10$)	$2.38e-7$
log_softmax naive	batched ($N=8, C=10$)	$4.77e-7$
log_softmax naive	wider- C stress ($N=2, C=128$)	$4.77e-7$
log_softmax naive	stability ($N=2$, logits ~ 80)	$2.38e-7$
GPU forward (E2E)	fixture 0 (MNIST idx 7)	$3.82e-6$
GPU forward (E2E)	fixture 1	$5.72e-6$
GPU forward (E2E)	fixture 2	$1.91e-6$
GPU forward (E2E)	fixture 3	$1.91e-6$
GPU forward (E2E)	fixture 4	$2.86e-6$
conv2d cuDNN	$N=2, 1 \rightarrow 10, 5 \times 5 @ 28 \times 28$	$1.91e-6$
conv2d cuDNN	$N=2, 10 \rightarrow 20, 5 \times 5 @ 12 \times 12$	$6.20e-6$
GPU forward 10k naive	all 10 000 MNIST (argmax 100%)	$1.91e-5$
GPU forward 10k tiled	all 10 000 MNIST (argmax 100%)	$1.91e-5$

TABLE II

PER-KERNEL CONVOLUTION THROUGHPUT ON RTX 4090 (MS PER LAUNCH, MEAN OVER 200 REPS WITH WARMUP). THE FINAL TWO COLUMNS ARE TILED AND cuDNN SPEEDUP RELATIVE TO THE NAIVE KERNEL.

Shape	Naive (ms)	Tiled (ms)	cuDNN (ms)	Tiled $\times$	cuDNN $\times$
conv1 ($N=1, 1 \rightarrow 10, 5^2 @ 28^2$)	0.0027	0.0030	0.0681	0.89	0.04
conv2 ($N=1, 10 \rightarrow 20, 5^2 @ 12^2$)	0.0123	0.0140	0.0855	0.88	0.14
conv1 ($N=64, 1 \rightarrow 10, 5^2 @ 28^2$)	0.0084	0.0101	0.0649	0.84	0.13
conv2 ($N=64, 10 \rightarrow 20, 5^2 @ 12^2$)	0.0288	0.0332	0.0661	0.87	0.44
synth ($N=1, 32 \rightarrow 64, 5^2 @ 64^2$)	0.0988	0.1197	0.0859	0.83	1.15

TABLE III

PER-KERNEL LINEAR / GEMM THROUGHPUT ON RTX 4090 AT LeNET-SCALE AND BATCH-64 SHAPES. PER-LAUNCH MEAN OVER 200 REPS WITH WARMUP. THE $6.43\times$ PEAK AT $N=1024$ IS SHOWN IN FIG. 4.

Shape	Naive (ms)	Tiled (ms)	Speedup
fc1 ($N=1, 320 \rightarrow 50$)	0.0107	0.0102	$1.06\times$
fc2 ($N=1, 50 \rightarrow 10$)	0.0025	0.0030	$0.83\times$
fc1 ($N=64, 320 \rightarrow 50$)	0.0375	0.0104	$3.60\times$

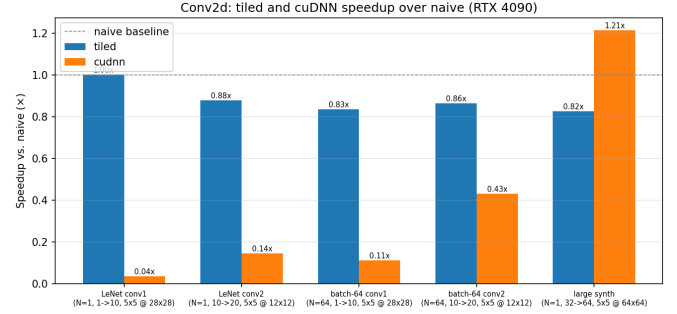


Fig. 2. Convolution speedup over the naive direct kernel on RTX 4090. cuDNN (IMPLICIT_PRECOMP_GEMM with FMA math to match FP32 numerics) pays a fixed per-launch overhead that dominates at LeNet scale and only crosses over on the larger $32 \rightarrow 64$ synthetic shape.

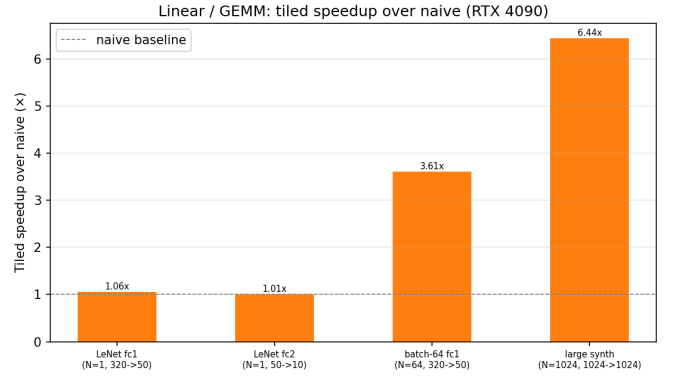


Fig. 3. Tiled-GEMM speedup over naive at each benchmarked linear shape. Both LeNet fc layers at batch 1 are break-even or worse (the output tile is at most 1×50 , most of each 16×16 block idles); batch-64 fc1 and the $1024 \times 1024 \times 1024$ synthetic case are where shared-memory reuse actually pays.

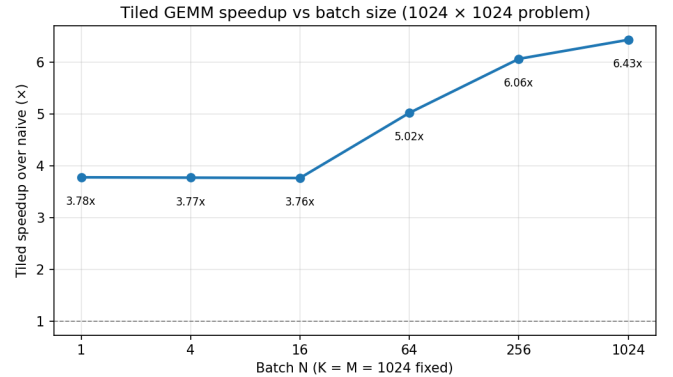


Fig. 4. Tiled-GEMM speedup over the naive kernel as the batch dimension N sweeps from 1 to 1024 at a fixed 1024×1024 weight shape. The plateau at $\sim 3.8\times$ for $N \leq 16$ reflects the regime in which the output M -tile is fully populated but only one K -panel is reused; the further climb to $6.43\times$ at $N=1024$ reflects full reuse along all three dimensions.

TABLE IV

END-TO-END LeNet FORWARD PASS ON RTX 4090: WEIGHTS AND INPUT H2D, ALL FIVE KERNELS, OUTPUT D2H. PER-CALL MEAN OVER 50 REPS WITH WARMUP. THROUGHPUT (IMG/S) IS FOR THE TILED PIPELINE.

Variant	Naive (ms)	Tiled (ms)	Speedup	img/s
forward $N=1$	0.1648	0.1362	$1.21\times$	7,342
forward $N=64$	0.4601	0.4351	$1.06\times$	147,100

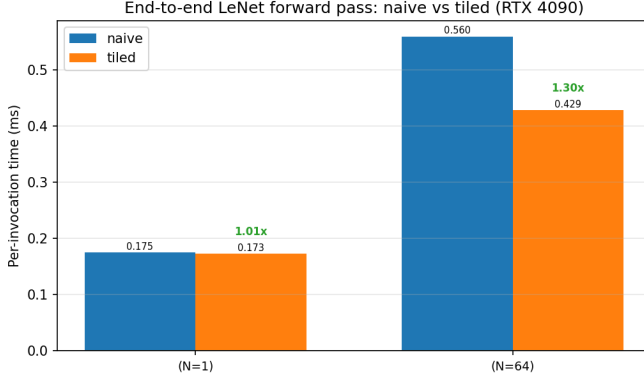


Fig. 5. End-to-end LeNet forward pass, naive vs tiled at batch 1 and batch 64 on RTX 4090. The tiled `fc1` win carries through to a $1.21\times$ end-to-end speedup at batch 1; the margin narrows at batch 64 because the tiled-conv regression is of comparable scale.

the working set is hot in the L1 and read-only caches. Tiling trades those already-cached global reads for shared-memory loads plus two `__syncthreads` per input-channel iteration, and at $K=5$ the arithmetic intensity is high enough that the kernel is already compute-bound, so the synchronization cost is a net loss. The regression persists on the off-model synthetic shape, suggesting the cross-over where shared-memory reuse actually pays is at filters large enough and channel counts high enough to exhaust the caches — well above anything LeNet inference touches.

The `cuDNN` column in Table II tells the same story from the opposite direction. `cuDNN` is pinned to `IMPLICIT_PRECOMP_GEMM` with `CUDNN_FMA_MATH` so that numerics match the FP32 direct-convolution reference (the default `CUDNN_DEFAULT_MATH` silently selects TF32 tensor cores on Ada, whose 10-bit mantissa produces $\sim 10^{-3}$ absolute error — larger than the project tolerance, and not comparable to the naive/tiled kernels). Under matched numerics, `cuDNN` is 7–25 $\times$ slower than the naive kernel at every LeNet shape; the per-launch descriptor build, workspace query, and implicit-GEMM dispatch cost on the order of $70\mu s$, which is pure overhead when the actual convolution takes single-digit microseconds. Only on the $32\rightarrow 64$, 64×64 synthetic shape does `cuDNN` overtake the naive kernel ($1.15\times$), and that crossover is consistent with the shared-memory argument above: once the working set is large enough that the naive kernel stops being L1-resident, any formulation that amortizes its dispatch overhead starts to win. Figure 2 visualizes all three

variants on a single axis.

The tiled GEMM tells a different story. At batch-64 `fc1` (64×320 times 320×50), a 16×16 tile fills with real work and the $16\times$ reuse factor along each input dimension shows up as a $3.60\times$ speedup. At $1024\times 1024\times 1024$ — a “real” GEMM workload — the tiled kernel is $6.43\times$ faster (Fig. 4). At batch-1 inference sizes the output tile is 1×10 or 1×50 , so most of each 16×16 block sits idle and the advantage disappears.

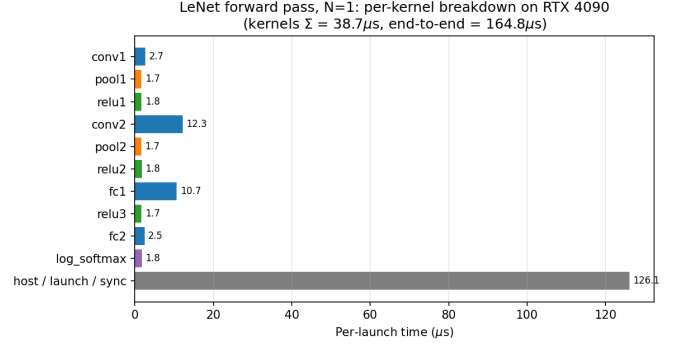


Fig. 6. Per-kernel contribution to the $N=1$ LeNet forward pass. The ten CUDA kernels together account for $\sim 39\mu s$; the remaining $\sim 126\mu s$ (the bottom bar) is host-side launch submission, CUDA runtime overhead, and synchronization on the final device-to-host copy. Conv2 and `fc1` dominate actual kernel work; every other kernel is within noise of the $\sim 1.7\mu s$ launch floor for a trivial CUDA kernel on this platform.

End-to-end the kernel-level story carries through weakly. Table IV measures the GPU-side pipeline only (weights are loaded and `cudaMemcpy`d once before timing starts, not on every invocation), so the reported times are dominated by the ten kernel launches plus their serialized activations. The tiled pipeline comes out $1.21\times$ faster at batch 1 and $1.06\times$ faster at batch 64: the tiled `fc1` win outweighs the small tiled-conv regression at LeNet scale, but not by a large margin. Shrinking the ten-launch overhead — for example by fusing `conv+bias+ReLU+pool` into a single kernel — is the biggest remaining end-to-end lever and is measured directly in Section IV-C.

C. Kernel Fusion: `conv+bias+ReLU+pool`

Section IV-B identifies launch overhead as the biggest end-to-end lever. The most obvious fusion target is the `conv` $\rightarrow$ `bias` $\rightarrow$ `ReLU` $\rightarrow$ `pool` block that runs twice in LeNet: three separate kernel launches become one, and the intermediate activation tensors are never materialized in global memory. Figure 7 sketches the memory-traffic story. The unfused path writes the full post-conv activation to DRAM, reads it back to apply ReLU in place, then reads again for pooling — three launch latencies plus two round-trips through global memory for data that is consumed immediately. The fused kernel keeps the input patch in dynamic shared memory, accumulates four conv outputs per thread (the receptive field of one pool output), applies the bias, ReLU, and 2×2 max in registers, and writes only the pool output to DRAM.

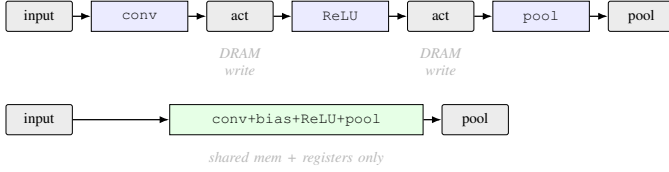


Fig. 7. Memory-traffic view of the fused stage. Top: three separate kernels, with the full post-conv and post-ReLU activations written to and read from DRAM between launches. Bottom: one kernel, with intermediate values kept in shared memory and registers; only the pool output hits DRAM.

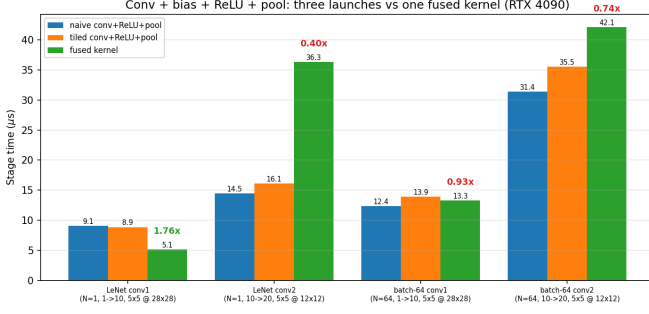


Fig. 8. Conv+bias+ReLU+pool stage time (μ s) on RTX 4090 across four shapes. Bars compare the three-kernel baseline (conv_naive $\rightarrow$ ReLU $\rightarrow$ max_pool2d, and the same with tiled conv) against the single fused kernel. Speedup over the naive three-kernel sequence is printed above the fused bar (green where the fused kernel wins, red where it loses).

The fused kernel uses a 4×4 pool-output block (16 threads producing a 4×4 pool tile, which corresponds to an 8×8 conv-output tile and a $(2 \cdot 4 + K_h - 1) \times (2 \cdot 4 + K_w - 1)$ dynamic shared-memory input patch per input channel). At LeNet $K_h = K_w = 5$ that is 576 B per input-channel iteration. Each thread holds four conv accumulators in registers, adds the bias, applies ReLU, and reduces the 2×2 window to one pool output. Correctness is validated at the project-wide 1×10^{-4} tolerance against the three-step CPU reference sequence (conv $\rightarrow$ ReLU $\rightarrow$ max_pool2d) on the two LeNet shapes, a batch-64 replica, and a non-pool-aligned synthetic shape; maximum observed error is 9.5×10^{-6} on conv2 at batch 2, well inside tolerance.

Figure 8 measures the stage three ways: three-kernel-sequence (naive conv), three-kernel-sequence (tiled conv), and the single fused kernel, all on the same random tensors with warmup and 200-rep `cudaEvent` amortization.

At LeNet conv1 and $N=1$ — the single-image predict regime — the fused kernel runs at 5.1μ s against 9.1μ s for the naive three-call sequence, a $1.76\times$ win. At LeNet conv2 and $N=1$, fusion regresses to $0.40\times$ of naive: the 4×4 pool output of conv2 launches only 20 blocks with 16 threads each (320 threads total) on 128 SMs, so the fused kernel under-subscribes the device and its heavier per-thread work (four conv accumulations per thread, plus the cooperative shared-memory load loop) dominates what was already a short naive-sequence stage. At batch 64 the fused kernel sits within 10–25% of the naive baseline on either shape; the

TABLE V
PER-KERNEL STATIC RESOURCE USAGE ON RTX 4090 (SM_89), FROM `NVCC -XPTXAS=-v`. DYNAMIC SHARED MEMORY REPORTED AT LEnET SHAPES (5×5 FILTER FOR CONV2D_TILED, $C=10$ FOR LOG_SOFTMAX). NO KERNEL SPILLS.

Kernel	Block	Regs	Static smem	Dyn. smem
conv2d_naive	16×16	39	0	0
conv2d_tiled	16×16	40	0	1600 B
conv_relu_pool_fused	4×4	39	0	576 B
linear_naive	256	38	0	0
linear_tiled	16×16	37	2048 B	0
max_pool2d_naive	16×16	20	0	0
relu_naive	256	8	0	0
log_softmax_naive	16	15	0	64 B

three launch latencies are now amortized across many output tiles, so the win from eliminating them is offset by the same small per-thread-overhead penalty. The takeaway is structural: fusion pays when launch overhead is the binding cost (small problems, tight dispatch loops) and costs when the fused kernel further starves the SMs of parallelism. At LeNet conv1 batch 1 both pressures line up in fusion’s favour, and it delivers a real end-to-end saving on exactly the shape a predict-style deployment runs.

D. Kernel Resource Usage and Occupancy

Nsight Compute was the intended profiling tool. In the RunPod container used for all measurements, Nsight Compute’s hardware performance-counter interface is blocked (`ERR_NVGPUCTRPERM`, gated by the `NVreg_RestrictProfilingToAdminUsers` driver parameter that is not configurable from an unprivileged container), so the timing-side substitute is the `cudaEvent` harness already described. Static kernel characterization, however, does not require counter access: `nvcc -Xptxas=-v` reports registers, shared memory and spill slots at compile time, and the launch bounds are fixed in the kernel sources. Table V collects both for each kernel as compiled for sm_89, and computes the resulting theoretical occupancy by hand against the RTX 4090’s limits (1536 threads/SM, 64K 32-bit registers/SM, ~ 99 KB shared memory/SM, 16 blocks/SM).

All six of the compute-heavy kernels reach the theoretical 100% occupancy ceiling: the regs-per-thread budget for a 256-thread block is $64K/256 = 256$ registers, and every kernel uses significantly fewer (≤ 40), so six full blocks (1536 threads, the thread-count ceiling) fit on each SM. No kernel spills and no kernel allocates a stack frame. This is an important reading of the throughput numbers in Section IV-B: the naive-vs-tiled delta at LeNet scale is not an occupancy story but a memory-access-pattern story, exactly as the per-kernel discussion there describes.

The two outliers are the 16-thread-block kernels: `log_softmax_naive` (block=16, one block per output row) and `conv_relu_pool_fused` (block= $4 \times 4 = 16$, one block per pool-output tile). With a 16-thread block, each

block occupies exactly one warp. The block-count ceiling of 16 blocks/SM is reached before either the thread-count or register ceiling, yielding 16 warps/48 max warps = 33% theoretical occupancy. Occupancy here is measured as active warps $\div$ maximum warps per SM, the standard CUDA definition. At LeNet’s batch-1 shapes both kernels complete in $\sim 1\text{--}3\mu\text{s}$ and occupancy is not the binding constraint on throughput.

E. Case Study: Matching Numerics Against cuDNN

Implementing the cuDNN baseline surfaced a numerical issue that is easy to miss and worth calling out. Ada Lovelace GPUs (compute capability 8.9, including the RTX 4090 used here) default to TF32 tensor cores for FP32 convolutions when cuDNN is invoked with the default math mode `CUDNN_DEFAULT_MATH`. TF32 keeps an FP32-wide exponent but truncates the mantissa to 10 bits, and on the conv2 shape this produced a maximum absolute error of $\sim 5.3 \times 10^{-3}$ against the direct-convolution CPU reference — an order of magnitude above the project’s 1×10^{-4} tolerance, and orders of magnitude above the FP32 summation-reorder noise floor observed elsewhere in this report. Similarly, the default cuDNN algorithm selector for the conv2 shape picks `WINOGRAD_NONFUSED`, whose algebraic transformation introduces a separate $\sim 10^{-3}$ error term. Getting cuDNN to produce numerics comparable to the direct-convolution kernels required two explicit pins: the math type set to `CUDNN_FMA_MATH` (via `cudaSetConvolutionMathType`) to force true FP32 fused-multiply-add, and the algorithm pinned to `IMPLICIT_PRECOMP_GEMM` to stay on direct convolution rather than Winograd. Under those pins cuDNN matches the naive kernel to $1.91\text{e-}6$ on conv1 and $6.20\text{e-}6$ on conv2. The practical lesson is that “use the vendor library” is not a numerical-accuracy free-win on modern GPUs — defaults are tuned for training throughput, not for parity with textbook floating-point semantics, and a meaningful throughput comparison requires matched math modes.

V. DISCUSSION AND SUMMARY

This project raised three observations worth spelling out.

First, the naive-first discipline paid off twice. The first time was the obvious win: because every kernel was validated against the CPU reference before it was integrated, the end-to-end forward pass came up green on the first attempt, with a maximum absolute error of 5.72×10^{-6} across all five fixtures and argmax matching on every image. No time was spent bisecting layer-to-layer logit divergence at 2am. The second win was less obvious: the per-kernel harness is what made it safe to add tiled variants and an additional benchmark surface without worrying about silent correctness regressions. The tiled kernels were written against the same interface, their unit tests exercised the same shapes, and promoting them into the end-to-end pipeline was a drop-in substitution.

Second, shared-memory tiling is not a universal optimization — it is a cache-reuse optimization, and LeNet inference

at the target scale is not cache-constrained. For 5×5 convolutions on 28×28 or 12×12 feature maps with ≤ 20 channels, the entire weight tensor is a few kilobytes and the input feature map for one block is a few kilobytes more; both sit comfortably in the RTX 4090’s L1, and the extra `__syncthreads` calls required by the tile load/compute split become the dominant cost. The tiled GEMM paints the opposite picture: at a $1024 \times 1024 \times 1024$ workload the tiled kernel is $6.43\times$ faster (Fig. 4). The same tiled GEMM at LeNet inference shapes with batch 1 barely breaks even because the output tile is 1×10 or 1×50 and most lanes of the block are idle. Batch 64 `fc1` sits between these regimes and shows a $3.6\times$ speedup. The lesson is not “tiling is bad at small shapes” — the kernel is correct and the tile geometry is reasonable — but that these shapes are just too small to exercise what tiling is actually good at. The exception that proves the rule is the fused `conv+bias+ReLU+pool` kernel at conv1 batch 1, where launch overhead is the binding cost and fusion delivers a $1.76\times$ end-to-end win (Section IV-C).

Third, the cuDNN comparison is best read as a calibration of where the direct-convolution formulation stops being the right choice, not as a throughput race this project could have won. cuDNN’s fixed per-launch cost ($\sim 70\mu\text{s}$ on this platform) exceeds the entire LeNet-scale convolution it is dispatching, so at the target shapes the hand-written direct kernel wins by $7\text{--}25\times$; but on the $32\rightarrow 64$, 64×64 synthetic shape cuDNN already overtakes the direct kernel, and at production-scale convolutions, the gap would keep widening. The reasons are structural, not a matter of tuning effort. cuDNN selects Winograd-family reformulations for 5×5 filters that cut arithmetic by a factor of two or more [3]; a direct kernel simply cannot recover that. On Ada Lovelace, cuDNN can also dispatch to tensor cores delivering roughly an order of magnitude more throughput than the FP32 CUDA cores used here — I intentionally disabled that path to preserve numerical parity (Section IV-E). On top of that, cuDNN ships hand-tuned kernels selected per $(C_{\text{in}}, C_{\text{out}}, H, W, K)$ shape tuple and can transpose between NCHW and NHWC to exploit 128-bit loads; this project uses a single generic tile geometry and stays in NCHW throughout to match PyTorch byte-for-byte. Each of those gaps is a project in itself. The comparison in Table II is therefore a reference point that establishes the crossover shape, not a target that was closed.

Beyond the throughput numbers, these additional investigations each produced a finding that was not obvious going in. The fused `conv+bias+ReLU+pool` kernel showed that launch-overhead elimination is a real and measurable lever at LeNet’s smallest shapes — $1.76\times$ at conv1 batch 1 — even when shared-memory tiling is not. The cuDNN numeric case study (Section IV-E) surfaced something less expected: Ada Lovelace’s `CUDNN_DEFAULT_MATH` silently selects TF32 tensor cores, whose 10-bit mantissa produced $\sim 5 \times 10^{-3}$ errors — an order of magnitude above the 1×10^{-4} tolerance — with no warning. Getting cuDNN to produce comparable numerics required pinning both the math type (`CUDNN_FMA_MATH`) and the algorithm (`IMPLICIT_PRECOMP_GEMM`), neither of

which is in the standard quickstart. It is a good reminder that “just use the vendor library” is not a numerical free-win on modern hardware. The engine matches PyTorch to 1.91×10^{-5} over 10,000 images with 100% argmax agreement and runs at 147,100 images/second at batch 64 on an RTX 4090 — fast enough to classify every frame of roughly 4,900 simultaneous 30 fps video streams in real time. The tiled kernels delivered $3.6\times$ over naive at batch-64 `fc1`. Going beyond the core scope, the fused kernel delivered a real $1.76\times$ win at `conv1` batch 1 where it mattered, and the cuDNN work uncovered the TF32 precision trap described above — the kind of thing that only shows up when you actually try to match numerics against the vendor library.

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Compute for all experiments was provided by RunPod (<https://runpod.io>) on an NVIDIA RTX 4090 GPU instance (CUDA 12.4, driver 550.127.05, cuDNN 9).

The ground-truth log-softmax outputs were generated by PyTorch [6] running my Project 5 weights in eval mode.

Anthropic’s Claude Code CLI was used as a development aid throughout this project. Specific uses include: code scaffolding and repository structure, kernel comment formatting, debugging assistance, rapid design prototyping, DevOps configuration for the RunPod compute environment, figure plotting with `matplotlib`, Beamer slide templating, and report prose editing for academic style and clarity. All architectural decisions, kernel designs, correctness methodology, benchmark measurements, and the analysis in this paper are my own; every line of CUDA was read, understood, and edited by me before it was committed.

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